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News Bot Reflection Journal

This was an interesting experience for our team to explore and implement different NLP techniques in one project altogether. We have learned different NLP concepts like TF-IDF, text preprocessing, POS Tagging, text classification, and sentiment analysis. We applied almost all concepts in this project, called a news bot intelligence system. which main purpose is to analyze news articles, categories, and recognize important information and grasp their emotional tone. This project makes us understand how NLP is used in research, news agencies, and businesses to process large amounts of information effectively.

We used the BBC news classification data set from KAGGLE. It has 1,490 articles and has five categories: technology, business, sports, politics, and entertainment. We used google colab as our cloud-based development environment and load our data in it as a first step, also identify data set structure using python and pandas then checked the data size, no of articles, missing values and length.

Next step was preprocessing in which we cleaned the data by removing needless characters, convert text to lower case removed stop words and applied lemmatization. TF-IDF used to convert 1,490 articles into numerical matrix containing 5,000 features.

Moving forward, another important part of this project was POS tagging and syntax parsing. POS tagging helped us identify nouns, verbs, adjectives, and other grammatical categories. Also, our assessment spots 43 different POS tags which made us understand the importance of grammatical patterns across different news categories. We used SpaCy for dependency parsing which determines how words are connected. In this project I found out that parsing is more challenging, especially when it comes to longer news articles.

Furthermore, VADER used to analyze the emotional tone of all 1,490 articles. The entertainment category had the highest sentiment score while politics had the lowest which showed different types of news have different emotional tones. We also performed names entity recognition (NER) which showed important information like locations, organizations, people, dates and money amount. Our system recognizes 42,301 entities mention across the articles also dates and people were among the most common entity types.

Finally , we trained and compared three machine learning algorithms: naive bayes, logistic Regression and support vector Machine (SVM). Naive bayes accuracy was 96.98%; logistic accuracy was 97.65% and SVM was 94.30%. Logistic regression performed best in all three and achieved 97.65% accuracy on our test dataset. This shows that it's not necessary that a complex model always performs better; simple models like logistic regression can bring better results too. We also examined precision, recall,confussion matrix and F! Score.

At last, we tested the Newsbot intelligence system using three new articles about a Microsoft acquistion, a basketball game and climate change. System identifies the categories, entities and sentiment successfully, but the confidence scores were not equally strong for example Microsoft article received 29.02% confidence, sports article received 92.7%, and climate change article received 45.41%. This information is interesting even though logistic regression achieved

97.65% accuracy during evaluation. When analyzing new articles, it was not equally confident. One more thing identified when looking at the results is that articles which contain information about multiple categories can make classification more difficult. For example, Microsoft acquisition can be related to both technology and business.

Overall, this project helped our team to combine all NLP concepts we learned so far and practice it through the real-world example.